Seat No.:_____

AHU ID No._____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

2023 Spring Semester

Applied Calculus III Midterm Exam

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	V	Total score
Score						
Grader						

I. Blank Fillings (5 items, 3 points each)

Score	

1. The angle between the line $\frac{x-1}{-1}=\frac{y}{1}=\frac{z+2}{2}$ and the plane $2x+y-z+1=0$ is _____.

2. A triangle in space with vertexes $A(1,-1,2), B(3,0,1)$ and $C(2,1,3)$, the area of the triangle is _____.

3. $\lim\limits_{(x,y)\to(1,0)} \frac{\sin xy}{y} =$ _____.

4. Let $f(x,y,z)=e^x yz^2$,where $z=z(x,y)$ is determined by implicit function $x+y+z+xyz=0$.Then $f_x'(0,1,-1)=$ _____.

5. The extremum of function $z=\sqrt{8-x^2+2x-y^2}$ is _____.

II. Choices (5 items, 2 points each)

Score	

6. The right line equation which is parallel to the line $L:\frac{x-1}{-1}=\frac{y-3}{3}=\frac{z+2}{-2}$ is( ).

A $\frac{x-1}{1}=\frac{y-2}{-3}=\frac{z}{2}$ B $\frac{x-1}{1}=\frac{y-2}{3}=\frac{z}{-2}$

C $\frac{x-1}{1}=\frac{y-2}{-3}=\frac{z}{-2}$ D $\frac{x-1}{1}=\frac{y-2}{3}=\frac{z}{2}$

7. The distance from the point(1,2,0) to the plane $2x-y+z=3$ is ( ).

A $\frac{3}{2}$ B $\frac{\sqrt{6}}{2}$ C $\sqrt{6}$ D 3

8. Let f be a binary function whose domain D contains some neighborhood of (a,b).Which is INCORRECT in the following statements? ( ).

A Both $f_x(a,b)$ and $f_y(a,b)$ exist if f is differentiable at (a,b).

B $f_{xy}(a,b)=f_{yx}(a,b)$ if f_{xy} and f_{yx} are continuous at (a,b).

C f is differentiable at (a,b) if f_x and f_y are continuous at (a,b).

D f is continuous at (a,b) if $f_x(a,b)$ and $f_y(a,b)$ exist.

9. Let $z=\frac{y}{f(u)}$,where $u=x^2-y^2$, $f(u)$ is derivable.Then $\frac{1}{x}\frac{\partial z}{\partial x}+\frac{1}{y}\frac{\partial z}{\partial y}$=( ).

A $\frac{y}{z^2}$ B $\frac{y}{z}$ C $\frac{z}{y^2}$ D $\frac{z}{y}$

10. Let $z=z(x,y)$ be implicit function determined by the equation $xyz+\sqrt{x^2+y^2+z^2}=\sqrt{2}$, then the differential $dz\,|_{(1,0,-1)}=$ ( ).

A $dx+\sqrt{2}dy$ B $-dx+\sqrt{2}dy$

C $-dx-\sqrt{2}dy$ D $dx-\sqrt{2}dy$

III. Calculations (5 items, 10 points each)

Score	

11. Let a plane pass through the point $(1,-2,1)$ and be perpendicular to the line of $\begin{cases} x+2y-3z+1=0 \\ 2x-y+z=0 \end{cases}$. Find the equation of the plane.

12. Let a line pass through the point $(1,1,-2)$, be intersected with the line $\frac{x}{2}=\frac{y}{1}=-z$ and perpendicular to the line $\frac{x-1}{3}=\frac{y}{2}=\frac{z+1}{1}$. Find the equation of the line.

13. Compute $\lim\limits_{(x,y)\to(0,0)} \frac{\sqrt{xy+1}-1}{e^{xy}-1}$.

14. Let $u = x^{\frac{y}{z}}$. Find du.

15. Suppose that $u = u(x, y), v = v(x, y)$ are determined by functions $\begin{cases} u^2 - e^v + x = 0 \\ e^u + v^2 - y = 0 \end{cases}$,

Find $\dfrac{\partial u}{\partial x}, \dfrac{\partial v}{\partial y}$.

IV. Proofs (2 items, 8 points each)

Score	

16. Prove that four points $A(1,0,1), B(4,4,6), C(2,2,3)$ and $D(9,12,15)$ are coplanar.

17. Let $z = f(xy, \frac{x}{y})$, where f has continuous second order partial derivatives. Prove that $\frac{\partial^2 z}{\partial x \partial y}$ is independent of f_{12}''.

V. Modelling Problem (9 points)

Score	

18. The intersecting curve of paraboloid $z = x^2 + y^2$ and the plane $x + y + z = 1$ is an ellipse. Find the minimal and maximal distance between original point and the ellipse.